

Reliability and Connection Establishment & Data Integrity and Ordering

Reliability and Connection Establishment

- TCP or UDP? **TCP**
- **Reasons?** Uses a 3-way handshake (SYN, SYN-ACK, ACK) before sending data, and formally closes the connection with FIN/ACK

Data Integrity and Ordering

- TCP or UDP? **TCP**
- **Reasons?** Uses Seq/Ack numbers to guarantee ordered delivery and retransmits any lost data

Use cases & Performance

Use cases

- **TCP** : Web browsing (HTTP/HTTPS), email, file transfer
- **UDP** : DNS, video/audio streaming, online gaming

Performance

- **TCP** : Slower due to handshake and acknowledgments, but reliable
- **UDP** : Faster and lighter due to simple header (no handshake), but not guaranteed